Interface Optimization: Technical Worksheet

Grade 10 · Worksheet · About 25 minutes

Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Interface optimization is improving a UI so users finish tasks faster, with fewer errors and less effort. Designers use **Fitts's Law**, **Hick's Law**, clear **hierarchy**, and **performance** improvements.

Do this:

1. **Name** one interface you use often that you would improve.

   _____

2. **Circle** the law about target size and distance: Fitts's Law · Newton's Law

Part 2 – Fitts and Hick (about 5 minutes)

Fitts's Law: big, close targets are faster to hit. **Hick's Law**: more choices make decisions slower.

> **Real-life scenario:** A page has 15 equal buttons. Users take longer to pick. The designer groups them into 4 clear sections.

Discuss with a partner or reflect in writing: How does reducing choices speed users up?

Do this: Circle the faster design.

Design	Faster?
Large Submit button near the form	Faster · Slower
Tiny Submit button far away	Faster · Slower
4 clear menu options	Faster · Slower
20 unlabeled options	Faster · Slower

Part 3 – Visual hierarchy (about 5 minutes)

Hierarchy shows what matters most using size, color, contrast, and position. The eye goes first to the biggest, boldest element.

> **Real-life scenario:** In a sign-up form, the main button is large and colorful; Cancel is a plain link.

Critical thinking: What happens if everything is the same size and color?

Word bank: *Size* · *Contrast* · *Position* · *Whitespace*

Do this: Choose the tool used.

Change	Hierarchy tool
Make title larger	_____
Use dark text on light background	_____
Leave space around a button	_____

Part 4 – Performance (about 5 minutes)

Speed is part of the interface. **Load time**, **lazy loading**, and **compressed images** make pages feel faster.

> **Real-life scenario:** A page loads in 8 seconds. After compressing images, it loads in 2.

Discuss with a partner or reflect in writing: Why do users leave slow pages?

Do this:

1. **Write** two ways to make a web page load faster.

   _____

2. **Circle** the faster approach: Compress images · Use huge uncompressed photos

Part 5 – Before and after (about 5 minutes)

Do this: Draw a cluttered screen and an optimized version with labels.

Write the metric you would measure to prove it is better.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I know **Fitts's** and **Hick's** laws.

☐ I can use **hierarchy**.

☐ I know load time affects users.

☐ I can compare two designs with a metric.

☐ I can propose optimizations.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Interface Optimization (Technical). Central topic: Optimizing interfaces. Main technology area: UI performance and design.